

Alphasyn EP 150

In accordance with Industrial Safety and Health Act

## Section 1. Chemical product and company identification

**Product name** Alphasyn EP 150  
**Code** 451075-FR01  
**SDS no.** 451075  
**Supplier** BP Korea Ltd.  
19F., 302, Teheran-ro, Gangnam-gu, Seoul, 06210  
Republic of Korea

Tel: +82 -1577-1904

**EMERGENCY TELEPHONE NUMBER** Carechem: +65 3158 1074 (24/7)

### Relevant identified uses of the substance or mixture and uses advised against

**Use of the substance/ mixture**  Gear lubricant.  
For specific application advice see appropriate Technical Data Sheet or consult our company representative.

## Section 2. Hazards identification

**GHS Classification** LONG-TERM (CHRONIC) AQUATIC HAZARD - Category 3

### GHS label elements, including precautionary statements

**Signal word** No signal word.  
**Hazard statements** H412 - Harmful to aquatic life with long lasting effects.

### Precautionary statements

**Prevention** P273 - Avoid release to the environment.  
**Response** Not applicable.  
**Storage** Not applicable.  
**Disposal** P501 - Dispose of contents and container in accordance with all local, regional, national and international regulations.

**Other hazards which do not result in classification**  Defatting to the skin.

## Section 3. Composition/information on ingredients

**Substance/mixture** Mixture  
Synthetic base stock. Proprietary performance additives.

### Hazardous ingredients

## Section 3. Composition/information on ingredients

| Ingredient name           | Synonym                                                                                                                                                                                                                                                                                                                                                                             | CAS number | %        |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| Amines, C12-14-tert-alkyl | (C12-14) tert-Alkyl primary amines; Primene 81-R; t-Alkyl-C12-14-primary amines; C12-14 tert-Alkylamines; Monoalkyl (or alkenyl, C5-28) amine; t-Alkyl (C12-C14) primary amines; tert-Alkyl(C=12-14) amines; (C12-14)tert-alkylamines; Tert-Alkyl (C12-C14), Amines; (C12-14)-tert-Alkyl-amine                                                                                      | 68955-53-3 | 0.132235 |
| (Z)-octadec-9-enylamine   | 9-Octadecen-1-amine, (9Z)-; Oleylamine; 9-Octadecen-1-amine, (Z)-; 9-Octadecenylamine, (Z)-; (9Z)-9-Octadecen-1-amine; Alamine 11; OLEAMINE; (Z)-9-Octadecen-1-amine; N-Alkyl [or alkenyl(C16-28)]-N, N-dialkyl(C1-5 or H) amine; N,N,N-Trialkyl (or alkenyl, at least one of the alkyl or alkenyl group is C8-24, others are H or C1-5) amine; Monoalkyl (or alkenyl, C5-28) amine | 112-90-3   | 0.026235 |

### Non-hazardous ingredients

| Ingredient name | Synonym       | CAS number    | %                |
|-----------------|---------------|---------------|------------------|
| Trade secret.   | Trade secret. | Trade secret. | 62.32            |
| Trade secret.   | Trade secret. | Trade secret. | 25               |
| Trade secret.   | Trade secret. | Trade secret. | 0.1 - 10         |
| Trade secret.   | Trade secret. | Trade secret. | 0.9275 - 1.06    |
| Trade secret.   | Trade secret. | Trade secret. | 0.3975 - 0.52973 |
| Trade secret.   | Trade secret. | Trade secret. | 0.265 - 0.3975   |
| Trade secret.   | Trade secret. | Trade secret. | 0.1325 - 0.265   |
| Trade secret.   | Trade secret. | Trade secret. | 0.1325 - 0.26235 |

There are no additional ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment and hence require reporting in this section.

Occupational exposure limits, if available, are listed in Section 8.

## Section 4. First aid measures

### Eye contact

In case of contact, immediately flush eyes with plenty of water for at least 15 minutes. Eyelids should be held away from the eyeball to ensure thorough rinsing. Check for and remove any contact lenses. Get medical attention.

### Skin contact

Wash skin thoroughly with soap and water or use recognised skin cleanser. Remove contaminated clothing and shoes. Wash clothing before reuse. Clean shoes thoroughly before reuse. Get medical attention if symptoms occur.

### Inhalation

If inhaled, remove to fresh air. Get medical attention if symptoms occur.

## Section 4. First aid measures

### Ingestion

Do not induce vomiting unless directed to do so by medical personnel. Never give anything by mouth to an unconscious person. If unconscious, place in recovery position and get medical attention immediately. Get medical attention if adverse health effects persist or are severe.

### Indication of immediate medical attention and special treatment needed, if necessary

### Specific treatments

No specific treatment.

### Notes to physician

Treatment should in general be symptomatic and directed to relieving any effects.

### Protection of first-aiders

No action shall be taken involving any personal risk or without suitable training. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation.

## Section 5. Firefighting measures

### Extinguishing media

#### Suitable

Use foam or all-purpose dry chemical to extinguish.

#### Not suitable

Do not use water jet.

#### Specific hazards arising from the chemical

Swarf fires - Neat metal working oils may fume, thermally decompose or ignite if they come into contact with red hot swarf. To minimise the generation of red hot swarf ensure that a sufficient flow of oil is correctly directed to the cutting edge of the tool to flood it throughout cutting operations. As an additional precaution swarf should be regularly cleared from the immediate area to prevent the risk of fire. Fire water contaminated with this material must be contained and prevented from being discharged to any waterway, sewer or drain. In a fire or if heated, a pressure increase will occur and the container may burst. This material is harmful to aquatic life with long lasting effects.

#### Hazardous thermal decomposition products

Combustion products may include the following:  
carbon oxides (CO, CO<sub>2</sub>) (carbon monoxide, carbon dioxide)  
sulphur oxides (SO, SO<sub>2</sub>, etc.)

#### Special protective equipment for fire-fighters

Fire-fighters should wear positive pressure self-contained breathing apparatus (SCBA) and full turnout gear.

#### Special protective actions for fire-fighters

No action shall be taken involving any personal risk or without suitable training. Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire.

## Section 6. Accidental release measures

#### Personal precautions, protective equipment and emergency procedures

Contact emergency personnel. No action shall be taken involving any personal risk or without suitable training. Evacuate surrounding areas. Keep unnecessary and unprotected personnel from entering. Do not touch or walk through spilt material. Avoid breathing vapour or mist. Provide adequate ventilation. Put on appropriate personal protective equipment. Floors may be slippery; use care to avoid falling.

#### Environmental precautions

Avoid dispersal of spilt material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air). Water polluting material. May be harmful to the environment if released in large quantities.

### Methods and material for containment and cleaning up

#### Small spill

Stop leak if without risk. Move containers from spill area. Absorb with an inert material and place in an appropriate waste disposal container. Dispose of via a licensed waste disposal contractor.

## Section 6. Accidental release measures

### Large spill

Stop leak if without risk. Move containers from spill area. Approach the release from upwind. Prevent entry into sewers, water courses, basements or confined areas. Contain and collect spillage with non-combustible, absorbent material e.g. sand, earth, vermiculite or diatomaceous earth and place in container for disposal according to local regulations. Contaminated absorbent material may pose the same hazard as the spilt product. Dispose of via a licensed waste disposal contractor.

## Section 7. Handling and storage

### Precautions for safe handling

#### Protective measures

Put on appropriate personal protective equipment (see Section 8). Do not ingest. Avoid contact with eyes, skin and clothing. Avoid breathing vapour or mist. Keep in the original container or an approved alternative made from a compatible material, kept tightly closed when not in use. Empty containers retain product residue and can be hazardous. Do not reuse container. Avoid contact of spilt material and runoff with soil and surface waterways. Concentrations of mist, fumes and vapours in enclosed spaces may result in the formation of explosive atmospheres. Excessive splashing, agitation or heating must be avoided. During metal working, solid particles from workpieces or tools will contaminate the fluid and may cause abrasions of the skin. Where such abrasions result in a penetration of the skin, first aid treatment should be applied as soon as reasonably possible. The presence of certain metals in the workpiece or tool, such as chromium, cobalt and nickel, can contaminate the metalworking fluid, as can bacteria, and as a result may induce allergic and other skin reactions, especially if personal hygiene is inadequate.

#### Advice on general occupational hygiene

Eating, drinking and smoking should be prohibited in areas where this material is handled, stored and processed. Wash thoroughly after handling. Remove contaminated clothing and protective equipment before entering eating areas. See also Section 8 for additional information on hygiene measures.

#### Conditions for safe storage, including any incompatibilities

Store in accordance with local regulations. Store in original container protected from direct sunlight in a dry, cool and well-ventilated area, away from incompatible materials (see Section 10) and food and drink. Keep container tightly closed and sealed until ready for use. Store and use only in equipment/containers designed for use with this product. Containers that have been opened must be carefully resealed and kept upright to prevent leakage. Do not store in unlabelled containers. Use appropriate containment to avoid environmental contamination.

#### Not suitable

☒ Prolonged exposure to elevated temperature

## Section 8. Exposure controls/personal protection

### Control parameters

| Ingredient name | Exposure limits |
|-----------------|-----------------|
| ☒ None.         |                 |

### Biological exposure indices

No exposure indices known.

Other ingredients including trade secret: not applicable

### Recommended monitoring procedures

If this product contains ingredients with exposure limits, personal, workplace atmosphere or biological monitoring may be required to determine the effectiveness of the ventilation or other control measures and/or the necessity to use respiratory protective equipment. Reference should be made to appropriate monitoring standards. Reference to national guidance documents for methods for the determination of hazardous substances will also be required.

## Section 8. Exposure controls/personal protection

### Appropriate engineering controls

All activities involving chemicals should be assessed for their risks to health, to ensure exposures are adequately controlled. Personal protective equipment should only be considered after other forms of control measures (e.g. engineering controls) have been suitably evaluated. Personal protective equipment should conform to appropriate standards, be suitable for use, be kept in good condition and properly maintained.

Your supplier of personal protective equipment should be consulted for advice on selection and appropriate standards. For further information contact your national organisation for standards.

Provide exhaust ventilation or other engineering controls to keep the relevant airborne concentrations below their respective occupational exposure limits.

The final choice of protective equipment will depend upon a risk assessment. It is important to ensure that all items of personal protective equipment are compatible.

### Environmental exposure controls

Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

### Personal protective equipment

#### Respiratory protection

In case of insufficient ventilation, wear suitable respiratory equipment.

For protection against metal working fluids, respiratory protection that is classified as "resistant to oil" (class R) or oil proof (class P) should be selected where appropriate. Depending on the level of airborne contaminants, an air-purifying, half-mask respirator (with HEPA filter) including disposable (P- or R-series) (for oil mists less than 50mg/m<sup>3</sup>), or any powered, air-purifying respirator equipped with hood or helmet and HEPA filter (for oil mists less than 125 mg/m<sup>3</sup>).

Where organic vapours are a potential hazard during metalworking operations, a combination particulate and organic vapour filter may be necessary.

The correct choice of respiratory protection depends upon the chemicals being handled, the conditions of work and use, and the condition of the respiratory equipment. Safety procedures should be developed for each intended application. Respiratory protection equipment should therefore be chosen in consultation with the supplier/manufacturer and with a full assessment of the working conditions.

#### Eye protection

Safety glasses with side shields.

#### Hand protection

Wear protective gloves if prolonged or repeated contact is likely. Wear chemical resistant gloves. Recommended: Nitrile gloves. The correct choice of protective gloves depends upon the chemicals being handled, the conditions of work and use, and the condition of the gloves (even the best chemically resistant glove will break down after repeated chemical exposures). Most gloves provide only a short time of protection before they must be discarded and replaced. Because specific work environments and material handling practices vary, safety procedures should be developed for each intended application. Gloves should therefore be chosen in consultation with the supplier/manufacturer and with a full assessment of the working conditions.

#### Skin protection

Use of protective clothing is good industrial practice.

Personal protective equipment for the body should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product.

Cotton or polyester/cotton overalls will only provide protection against light superficial contamination that will not soak through to the skin. Overalls should be laundered on a regular basis. When the risk of skin exposure is high (e.g. when cleaning up spillages or if there is a risk of splashing) then chemical resistant aprons and/or impervious chemical suits and boots will be required.

## Section 8. Exposure controls/personal protection

### Hygiene measures

Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Appropriate techniques should be used to remove potentially contaminated clothing. Wash contaminated clothing before reusing. Ensure that eyewash stations and safety showers are close to the workstation location.

## Section 9. Physical and chemical properties

The conditions of measurement of all properties are at standard temperature and pressure unless otherwise indicated.

### Appearance

#### Physical state

Liquid.

#### Colour

Yellow. [Light]

#### Odour

Not available.

#### Odour threshold

Not available.

#### pH

☒ Not applicable.

#### Melting/freezing point

Not available.

#### Boiling point, initial boiling point, and boiling range

Not available.

#### Flash point

Closed cup: >180°C (>356°F) [Pensky-Martens]  
Open cup: >200°C (>392°F) [Cleveland]

#### Evaporation rate

Not available.

#### Flammability (solid, gas)

Not available.

Not applicable. Based on - Physical state

#### Lower and upper explosive (flammable) limits

Not available.

#### Vapour pressure

Not available.

| Ingredient name | Vapour Pressure at 20°C |     |        | Vapour pressure at 50°C |     |        |
|-----------------|-------------------------|-----|--------|-------------------------|-----|--------|
|                 | mm Hg                   | kPa | Method | mm Hg                   | kPa | Method |
|                 |                         |     |        |                         |     |        |

### Solubility(ies)

| Media                                     | Result      |
|-------------------------------------------|-------------|
| <input checked="" type="checkbox"/> Water | Not soluble |

#### Vapour density

Not available.

#### Relative density

Not available.

#### Density

<1000 kg/m<sup>3</sup> (<1 g/cm<sup>3</sup>) at 15°C

#### Partition coefficient: n-octanol/water

☒ Not applicable.

#### Auto-ignition temperature

Not available.

#### Decomposition temperature

Not available.

#### Viscosity

☒ Kinematic: 150 mm<sup>2</sup>/s (150 cSt) at 40°C  
Kinematic: 19.8 mm<sup>2</sup>/s (19.8 cSt) at 100°C (ASTM D 445)

#### Molecular weight

Not applicable as it is a mixture

### Particle characteristics

#### Median particle size

☒ Not applicable.

## Section 10. Stability and reactivity

|                                           |                                                                                                                                                                         |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Chemical stability</b>                 | The product is stable.                                                                                                                                                  |
| <b>Possibility of hazardous reactions</b> | Under normal conditions of storage and use, hazardous reactions will not occur.<br>Under normal conditions of storage and use, hazardous polymerisation will not occur. |
| <b>Conditions to avoid</b>                | Avoid all possible sources of ignition (spark or flame).                                                                                                                |
| <b>Incompatible materials</b>             | Reactive or incompatible with the following materials: oxidising materials.                                                                                             |
| <b>Hazardous decomposition products</b>   | Under normal conditions of storage and use, hazardous decomposition products should not be produced.                                                                    |

## Section 11. Toxicological information

**Information on likely routes of exposure**       Routes of entry anticipated: Dermal, Inhalation, Eyes.

### Acute toxicity

**Inhalation**      Vapour inhalation under ambient conditions is not normally a problem due to low vapour pressure.

**Ingestion**      No known significant effects or critical hazards.

**Skin contact**      Defatting to the skin. May cause skin dryness and irritation.

**Eye contact**      No known significant effects or critical hazards.

### **Symptoms related to the physical, chemical and toxicological characteristics**

**Inhalation**      No specific data.

**Ingestion**      No specific data.

**Skin**      Adverse symptoms may include the following:  
irritation  
dryness  
cracking

**Eyes**      No specific data.

### **Delayed and immediate effects as well as chronic effects from short and long-term exposure**

| Product/ingredient name | Test | Species | Result | Exposure | Remarks |
|-------------------------|------|---------|--------|----------|---------|
|-------------------------|------|---------|--------|----------|---------|

 Not available.

### **Irritation/Corrosion**

| Product/ingredient name | Test authority / Test number | Species | Route / Result | Conc. | Remarks |
|-------------------------|------------------------------|---------|----------------|-------|---------|
|-------------------------|------------------------------|---------|----------------|-------|---------|

 Not available for product and all ingredients.

**Skin corrosion or irritation**      Not available for product and all ingredients.

**Serious eye damage/eye irritation**      Not available for product and all ingredients.

**Respiratory Irritation**      Not available for product and all ingredients.

### **Sensitisation**

**Respiratory Sensitisation**      Not available for product and all ingredients.

**Skin Sensitisation**      Not available for product and all ingredients.



## Section 11. Toxicological information

| Product/ingredient name                        | Route of exposure | Species | Result | Remarks |
|------------------------------------------------|-------------------|---------|--------|---------|
| Not available for product and all ingredients. |                   |         |        |         |

### CMR - ISHA Article 42 Public Notice No 2016-41 Occupational Exposure Limits

| Product/ingredient name                        | CAS number | Classification |
|------------------------------------------------|------------|----------------|
| Not available for product and all ingredients. |            |                |

### Carcinogenicity

Not available for product and all ingredients.

### Germ cell mutagenicity

| Product/ingredient name                        | Test | Experiment | Result | Remarks |
|------------------------------------------------|------|------------|--------|---------|
| Not available for product and all ingredients. |      |            |        |         |

### Reproductive toxicity

| Product/ingredient name                        | Test detail                                       | Species | Exposure | Developmental toxin | Maternal toxicity | Fertility | Remarks |
|------------------------------------------------|---------------------------------------------------|---------|----------|---------------------|-------------------|-----------|---------|
| Not available for product and all ingredients. |                                                   |         |          |                     |                   |           |         |
| <b>Teratogenicity</b>                          | No known significant effects or critical hazards. |         |          |                     |                   |           |         |
| <b>Developmental effects</b>                   | No known significant effects or critical hazards. |         |          |                     |                   |           |         |
| <b>Fertility effects</b>                       | No known significant effects or critical hazards. |         |          |                     |                   |           |         |

### Specific target organ toxicity (single exposure)

| Name                      | Category   | Route of exposure | Target organs                |
|---------------------------|------------|-------------------|------------------------------|
| Amines, C12-14-tert-alkyl | Category 3 | -                 | Respiratory tract irritation |
| (Z)-octadec-9-enylamine   | Category 3 | -                 | Respiratory tract irritation |

### Specific target organ toxicity (repeated exposure)

| Name                    | Category   | Route of exposure | Target organs |
|-------------------------|------------|-------------------|---------------|
| (Z)-octadec-9-enylamine | Category 2 | -                 | -             |

### Potential chronic health effects

|                        |                                                   |
|------------------------|---------------------------------------------------|
| <b>General</b>         | No known significant effects or critical hazards. |
| <b>Carcinogenicity</b> | No known significant effects or critical hazards. |
| <b>Mutagenicity</b>    | No known significant effects or critical hazards. |

### Aspiration hazard

| Name                    | Result                         |
|-------------------------|--------------------------------|
| (Z)-octadec-9-enylamine | ASPIRATION HAZARD - Category 1 |

**Other information** Not available.



## Section 12. Ecological information

### Ecotoxicity

This material is harmful to aquatic life with long lasting effects.

### Persistence/degradability

Not expected to be rapidly degradable.

### Mobility in soil

Spillages may penetrate the soil causing ground water contamination.

### Bioaccumulative potential

This product is not expected to bioaccumulate through food chains in the environment.

### Other adverse effects

No known significant effects or critical hazards.

### Other ecological information

Spills may form a film on water surfaces causing physical damage to organisms. Oxygen transfer could also be impaired.

## Section 13. Disposal considerations

### Disposal methods

The generation of waste should be avoided or minimised wherever possible. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Waste should not be disposed of untreated to the sewer unless fully compliant with the requirements of all authorities with jurisdiction. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible.

### Disposal precautions

This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Avoid dispersal of spilt material and runoff and contact with soil, waterways, drains and sewers.

## Section 14. Transport information

|                                      | IMDG           | IATA           |
|--------------------------------------|----------------|----------------|
| <b>A. UN number</b>                  | Not regulated. | Not regulated. |
| <b>B. UN proper shipping name</b>    | -              | -              |
| <b>C. Transport hazard class(es)</b> | -              | -              |
| <b>D. Packing group</b>              | -              | -              |
| <b>E. Environmental hazards</b>      | No.            | No.            |
| <b>F. Additional information</b>     | -              | -              |

### Special precautions for user

Not available.

## Section 15. Regulatory information

### Regulation according to ISHA

ISHA article 117 (Harmful substances prohibited from manufacture) None of the components are listed.

ISHA article 118 (Harmful substances requiring permission) None of the components are listed.

### Exposure Limits of Chemical Substances and Physical Factors

The following components have an OEL:

ISHA Enforcement Regs Annex 19 (Exposure standards established for harmful factors) The following components are listed: methanol, ethylene oxide

ISHA Enforcement Regs Annex 21 (Harmful factors subject to Work Environment Measurement) None of the components are listed.

ISHA Enforcement Regs Annex 22 (Harmful Factors Subject to Special Health Check-up) None of the components are listed.

Standard of Industrial Safety and Health Annex 12 (Hazardous substances subject to control) None of the components are listed.

### Regulation according to Chemicals Control Act

Article 20 Toxic Chemicals (K-Reach Article 20) Not applicable

Article 18 Prohibited (K-Reach Article 27) None of the components are listed.

Article 20 Restricted (K-Reach Article 27) None of the components are listed.

CCA Article 11 (TRI) None of the components are listed.

CCA Article 39 (Accident Precaution Chemicals) None of the components are listed.

Dangerous Materials Safety Management Act  
**Class:** Class 4 - Flammable Liquid  
**Item:** 6. Class 4 petroleum  
**Threshold:** 6000 L  
**Danger category:** III  
**Signal word:** Contact with sources of ignition prohibited

Wastes regulation Designated Waste

### Regulation according to other foreign laws

Australia inventory (AIIIC) All components are listed or exempted.

Canada inventory All components are listed or exempted.

China inventory (IECSC) All components are listed or exempted.

REACH Status The company, as identified in Section 1, sells this product in the EU in compliance with the current requirements of REACH.

Japan inventory (CSCL) All components are listed or exempted.

## Section 15. Regulatory information

|                                          |                                        |
|------------------------------------------|----------------------------------------|
| <b>Korea inventory (KECI)</b>            | All components are listed or exempted. |
| <b>Philippines inventory (PICCS)</b>     | All components are listed or exempted. |
| <b>Taiwan inventory (TCSI)</b>           | All components are listed or exempted. |
| <b>United States inventory (TSCA 8b)</b> | All components are active or exempted. |

## Section 16. Other information

### History

#### Source of Information

Sources of key data used to compile the Safety Data Sheet: Hazard assessment review data, toxicological reviews, and product physical properties; component supplier hazard communication data; and other publically available resources.

#### Date first prepared

15/07/2013

#### Number of revisions and date of last revision

10 07/05/2024.

#### Prepared by

Product Stewardship

#### Key to abbreviations

AMP = Acceptable Maximum Peak  
ACGIH = American Conference of Governmental Industrial Hygienists, an agency that promulgates exposure standards.  
ADG = Australian Code for the Transport of Dangerous Goods by Road and Rail  
ADG Code = Australian Code for the Transport of Dangerous Goods by Road and Rail  
CAS Number = Chemical Abstracts Service Registry Number  
HAZCHEM Code = Emergency action code of numbers and letters which gives information to emergency services. Its use is required by the ADG Code for Dangerous Goods in bulk.  
ICAO = International Civil Aviation Organization.  
IATA = International Air Transport Association, the organization promulgating rules governing shipment of goods by air.  
IMDG = International Maritime Organization Rules, rules governing shipment of goods by water.  
IP 346 = A chemical screening assay for dermal toxicity. The European Commission has recommended that Method IP 346 be used as the basis for labelling certain lubricant oil base stocks for carcinogenicity. The EU Commission has stipulated that the classification as a carcinogen need not apply if it can be shown that the substance contains less than 3% DMSO extract as measured by IP 346. (See Note L, European Commission Directive 67/548/EEC as amended and adapted.) DMSO is a solvent.  
NOHSC = National Occupational Health & Safety Commission, Australia  
REACH = Registration, Evaluation, Authorisation and Restriction of Chemicals Regulation [Regulation (EC) No. 1907/2006]  
TWA = Time weighted average  
STEL = Short term exposure limit  
UN Number = United Nations Number, a four digit number assigned by the United Nations Committee of Experts on the Transport of Dangerous Goods.  
TCCA = Toxic Chemical Control Act  
GHS = Global Harmonized System  
ISHA = Industrial Safety and Health Act  
Varies = may contain one or more of the following 64741-88-4, 64741-89-5, 64741-95-3, 64741-96-4, 64742-01-4, 64742-44-5, 64742-45-6, 64742-52-5, 64742-53-6, 64742-54-7, 64742-55-8, 64742-56-9, 64742-57-0, 64742-58-1, 64742-62-7, 64742-63-8, 64742-65-0, 64742-70-7, 72623-85-9, 72623-86-0, 72623-87-1

Indicates information that has changed from previously issued version.

## Section 16. Other information

### [Notice to reader](#)

All reasonably practicable steps have been taken to ensure this data sheet and the health, safety and environmental information contained in it is accurate as of the date specified below. No warranty or representation, express or implied is made as to the accuracy or completeness of the data and information in this data sheet.

The data and advice given apply when the product is sold for the stated application or applications. You should not use the product other than for the stated application or applications without seeking advice from BP Group.

It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. The BP Group shall not be responsible for any damage or injury resulting from use, other than the stated product use of the material, from any failure to adhere to recommendations, or from any hazards inherent in the nature of the material. Purchasers of the product for supply to a third party for use at work, have a duty to take all necessary steps to ensure that any person handling or using the product is provided with the information in this sheet. Employers have a duty to tell employees and others who may be affected of any hazards described in this sheet and of any precautions that should be taken. You can contact the BP Group to ensure that this document is the most current available. Alteration of this document is strictly prohibited.